

Formal Response to the Referee Report on the Split-Forest Automata Proof

Prepared review and repair memorandum

May 31, 2026

Overall assessment

I agree with the referee’s central criticism. The previous automata proof had the right high-level architecture, but the finite abstraction layer was not adequately justified. In particular, the proof attempted to recover a global RCC5 labelling from local pair/triple transformer states. The missing step was a confluence theorem saying that initialising at an ancestor and pushing endpoints down gives the same pair relation as initialising at the least common interface. That theorem had not been proved, and the local transition checks did not enforce it.

The corrected proof therefore removes the transformer construction. The replacement is a patchwork-bag abstraction: each finite bag carries a complete atomic RCC5 network, adjacent bags agree on overlaps, and global RCC5 consistency follows from the patchwork property of RCC5. Equality is no longer handled by quotienting repeated blocking copies; it is handled as exact occurrence identity in a tree decomposition. The automaton is also made explicitly two-way, and universal propagation is checked with exact relation-vector shadows rather than a may-analysis.

The repaired proof is supplied separately as `split_forest_patchwork_automata_repaired.tex/pdf`.

Response by issue

Issue	Referee concern	Repair
C1	The relation $\rho_T(\xi, \eta)$ was computed by route-dependent pair transformers, and triple projections need not agree with independently computed pair representatives.	The transformer layer is deleted. The new proof uses finite complete RCC5 bag networks and the RCC5 patchwork property. Global relations are supplied by an amalgamated complete network, not by an interface route.
C2	Multivalued composition allows local declarations that are legal in isolation but not globally coherent; diagnostic examples such as C_{split} are not forced to be checked in a common interface.	All finite bags are complete networks, and all finite tree-shaped unions of bags are amalgamated by patchwork. Thus forced exclusions such as $\text{Comp}(\text{PPI}, \text{PP}) = \{\text{EQ}, \text{PP}, \text{PPI}, \text{PO}\}$ are enforced in every global completion.

C3	Equality-port congruence was a global for-all- ζ condition reduced to finite frontier agreement; split copies in different trees might never be checked together.	The revised proof removes nontrivial semantic EQ quotienting. Multiple manifestations of one semantic occurrence are connected by overlap maps in a tree decomposition. Strong EQ is identity of overlap-equivalence classes; distinct local ports are never labelled EQ.
M1	The side-context width bound was a counting heuristic.	The proof now uses an effective type/mosaic enumeration bound. Saturated context shapes are classified up to finite RCC5 bag network, type assignment, overlap pattern, witness declarations, and shadow summaries. A numerical recurrence is not claimed.
M2	Shadow-class finiteness was asserted.	A shadow is explicitly defined as a finite profile $(t, \vec{r}, \Omega) \in \mathbf{Typ} \times \mathbf{Rel}^K \times 2^{\mathbf{Rel} \times \mathbf{Cl}(C_0)}$. The update into adjacent bags depends only on this profile and the adjacent finite bag network, so equal shadows are a congruence.
M3	One-way sufficiency depended on unproved lossless summaries.	The decision layer is now a two-way alternating parity tree automaton. The proof no longer requires a one-way lossless ancestor summary.
M4	Universal propagation used a may-analysis that might drop obligations when an intermediate relation leaves and later re-enters the guarded relation.	The automaton carries exact relation-vector shadows. If the recorded relation from the source shadow to a newly introduced target is R , then every obligation $\forall R.D$ forces D at that target. Obligations are not pruned by intermediate may-steps.
m3	Equality-mediated eventualities were informal.	Requests are attached to concrete overlap-equivalence classes. Since equality is identity, a request cannot be deferred forever by bouncing between distinct equality mates. The generalized Buchi request condition is converted to a finite parity index.

What remains from the original split-forest abstraction

The repaired proof does not abandon the split-forest idea. It keeps the split forest as the semantic normal form:

$$\text{witness thinning} \rightarrow \text{generated containment covers} \rightarrow \text{saturated side contexts.}$$

What changes is the adequacy proof for the finite abstraction. The old proof tried to hand-build a global relation by local transformers. The repaired proof represents the normal form by finite patchwork bags and lets the RCC5 patchwork theorem provide global amalgamation. The automaton is used only to decide the existence of such a regular patchwork-bag presentation and to manage infinite eventualities.

Remaining external input

The repaired proof is self-contained except for two standard background results:

1. the patchwork property for complete atomic RCC5 networks;
2. decidability of emptiness for finite two-way alternating parity tree automata.

If a completely self-contained manuscript is desired, the RCC5 patchwork theorem should be proved in an appendix or cited precisely. The referee explicitly recommended this patchwork route, and it is now the central replacement for the former unproved transformer-coherence lemma.

Bottom line

The referee's report identified a genuine defect in the abstraction-adequacy layer. The repaired proof addresses it by replacing route-dependent transformer coherence with patchwork amalgamation, exact tree-decomposition identity, finite shadow profiles, and a two-way parity automaton. This materially strengthens the proof and localizes the remaining mathematical dependence to the standard RCC5 patchwork theorem and the standard two-way parity-automaton emptiness theorem.